

Empirical Methods for Policy Evaluation

Matteo Bobba

Toulouse School of Economics (TSE)

TSE PhD Program (MRes)

Fall 2026

Part 1: Randomized Experiments and Dynamic Latent Factor Models

① Randomized control trials (RCTs)

- ⇒ SUTVA and assignment mechanisms
- ⇒ Randomization inference
- ⇒ Randomization designs
- ⇒ Stratified RCTs and Clustered RCTs

② Dynamic Latent Factor Models

- ⇒ Cunha-Nielsen-Williams (2021, ARE)

③ The production of human capital in the early years

- ⇒ [Attanasio-Cattan-Fitzsimons-Meghir-Rubio \(AER, 2020\)](#)

SUTVA and assignment mechanisms

Causal Inference as a Missing Data Problem

- Population of units, indexed by $i = 1, \dots, N$
- Treatment indicator W_i taking values 0 and 1
- For $i \in \{1, \dots, N\} \exists$ one realized outcome and one missing potential outcome

$$Y_i^{\text{obs}} = Y_i(W_i) = \begin{cases} Y_i(0) & \text{if } W_i = 0 \\ Y_i(1) & \text{if } W_i = 1 \end{cases}$$

$$Y_i^{\text{miss}} = Y_i(1 - W_i) = \begin{cases} Y_i(1) & \text{if } W_i = 0 \\ Y_i(0) & \text{if } W_i = 1 \end{cases}$$

- Unit-level causal effect $Y_i(1) - Y_i(0)$ is unobserved

The Stable Unit Treatment Value Assumption (SUTVA)

- We will generally need to predict, or impute, the missing potential outcome
- Denote $W_{-i} = (W_j)_{j \neq i}$, SUTVA requires that

$$(Y_i(1), Y_i(0)) \perp W_{-i}$$

1 No interference

⇒ Potential outcomes for unit i do not vary with the treatment status of any other j

2 Scale Invariance

⇒ There are no different forms of each treatment, which lead to $\neq$ potential outcomes

Sources of Possible Violations of SUTVA

1 Spillovers and equilibrium effects. E.g.:

- ⇒ Fertilizer in one plot may affect yields in contiguous plots
- ⇒ Immunization efficacy may depend on the number of people immunized
- ⇒ Prob(job) after training may be affected by the number of people trained

2 Hidden variations of treatments. E.g.:

- ⇒ Endogenous compliance to treatment assignment
- ⇒ Unobserved differences in the method of administering the treatment

Assignment Mechanism

- N units, set $\mathbb{W} = \{0, 1\}^N$ of N -vectors with all elements equal to 0 or 1
- The **assignment mechanism** is a function $P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) \in [0, 1]$ such that

$$\sum_{\mathbf{W} \in \{0, 1\}^N} P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = 1$$

⇒ The unit-level **assignment probability** is

$$p_i(\mathbf{Y}(0), \mathbf{Y}(1)) = \sum_{\mathbf{W}: W_i=1} P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1))$$

⇒ The **propensity score** is

$$e(x) = \frac{1}{N(x)} \sum_{i: X_i=x} p_i(\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1))$$

Example of Assignment Mechanism 1

- Two units, and so (2^2) possible values for $\mathbf{W}$

$$\mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

- $\Rightarrow$ The assignment mechanism is equal to $P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = \frac{1}{4}$
- $\Rightarrow$ Unit assignment probability $p_i = p = \frac{1}{2}$ for $i = 1, 2$
- $\Rightarrow$ No covariates, and so propensity score $e = p = \frac{1}{2}$

Example of Assignment Mechanism 2

- Two units where only assignments with one treated and one control unit

⇒ The assignment mechanism is

$$P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = \begin{cases} 1/2 & \text{if } \mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} \\ 0 & \text{if } \mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} \end{cases}$$

⇒ As before, $e = p = \frac{1}{2}$

Regular Assignment Mechanisms

1 Individualistic

$$p_i(\mathbf{Y}(0), \mathbf{Y}(1)) = q(Y_i(0), Y_i(1)), q(\cdot) \in [0, 1]$$

2 Probabilistic

$$0 < p_i(\mathbf{Y}(0), \mathbf{Y}(1)) < 1$$

3 Unconfounded

$$P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = P(\mathbf{W}|\mathbf{Y}'(0), \mathbf{Y}'(1)) = P(\mathbf{W})$$

Randomized Experiments and Observational Studies

- Individualistic + unconfounded implies

$$P(\mathbf{Y}(0), \mathbf{Y}(1)) = c \cdot \prod_{i=1}^N q^{W_i} (1 - q)^{1 - W_i}$$

⇒ The constant c ensures that the probabilities add to unity

⇒ In this case, $p_i = e = q$

- A regular assignment mechanism with **known and controlled** $q(\cdot)$ is a randomized experiment
- When the **functional form of the assignment mechanism is unknown** we have an observational study

Randomization Inference

Two Approaches for Inference (Not Just in RCTs)

① Asymptotic inference (model-based)

- ⇒ (semi-) **Parametric** models for the conditional mean of observed outcomes
- ⇒ **Observed outcomes** vary through random sampling from a population of units
- ⇒ Compute the distribution of the test statistic through **large-sample properties**

② Randomization inference (design-based)

- ⇒ **Nonparametric**; i.e. no restrictions on the distribution of the potential outcomes
- ⇒ **Treatment assignment** is the sole source of randomness; observed outcomes are fixed
- ⇒ The **assignment mechanism** determines the distribution of the test statistic

Illustrative Example

Unit	Potential Outcomes				
	Cough Frequency (cfa)		Observed Variables		
	$Y_i(0)$	$Y_i(1)$	W_i	X_i (cfp)	Y_i^{obs} (cfa)
1	?	3	1	4	3
2	?	5	1	6	5
3	?	0	1	4	0
4	4	?	0	4	4
5	0	?	0	1	0
6	1	?	0	5	1

The Sharp Null

- The p -value for the **sharp** null hypothesis that the treatment had no effect on cough frequency is

$$H_0 : Y_i(0) = Y_i(1) \forall i = 1, \dots, 6.$$

$\Rightarrow$ The average null $\mathbb{E}[Y_i(0) - Y_i(1)] = 0$ is **weaker than the sharp null** hypothesis

- Under H_0 , all missings in **potential outcomes inferred from observed outcomes**
- One **test statistics** from example above is

$$\begin{aligned} T(W, Y^{\text{obs}}) &= | \bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}} | \\ &= | (Y_1^{\text{obs}} + Y_2^{\text{obs}} + Y_3^{\text{obs}})/3 - (Y_4^{\text{obs}} + Y_5^{\text{obs}} + Y_6^{\text{obs}})/3 | \\ &= | 8/3 - 5/3 | = 1.00 \end{aligned}$$

Treatment Assignments Generate Test Distribution

- We can re-do this for $\binom{6}{3} = 20$ **permutations of treatment assignments**
⇒ E.g. instead of $W^{\text{obs}} = (1, 1, 1, 0, 0, 0)$ take $\tilde{W} = (0, 1, 1, 0, 0, 1)$
- **No change in observed outcomes** since $Y_i(0) = Y_i(1) = Y_i^{\text{obs}}$ under H_0
- The value of the test statistic for $\tilde{W}$ is

$$\begin{aligned} T(\tilde{W}, Y^{\text{obs}}) &= | (Y_2^{\text{obs}} + Y_3^{\text{obs}} + Y_6^{\text{obs}})/3 - (Y_1^{\text{obs}} + Y_4^{\text{obs}} + Y_5^{\text{obs}})/3 | \\ &= | 6/3 - 7/3 | = 0.33 \end{aligned}$$

Exact p -values

- Calculate the value of the statistic for each assignment vector
 - ⇒ In previous example, each assignment vector has prior probability=1/20
- How unusual is $T(W, Y^{\text{obs}}) = 1.00$ under the sharp null hypothesis?
- There are 16/20 assignments with $T(\tilde{W}, Y^{\text{obs}}) > T(W, Y^{\text{obs}})$: $p\text{-value} = 0.80$
- In this example, the observed difference could well be due to chance

Approximate p -values

- Recall that the number of distinct values of the treatment vector is $\binom{N_c + N_t}{N_t}$
 - ⇒ For instance, if $N = 100$ and $q = 0.5$ then $\dim(\mathbb{W}^+) = e^{29}$
- We thus need to rely on **numerical approximations** to calculate the p -value
 - ⇒ Draw an N -dimensional vector with N_c zeros and N_t ones from $\mathbb{W}^+$
 - ⇒ Repeat this process $K - 1$ times and approximate the p -value by:

$$\hat{p} = \frac{1}{K} \sum_{k=1}^K \mathbf{1}_{T^{\text{dif},k} \geq T^{\text{dif},\text{obs}}}$$

- ⇒ With $K > 1,000$ each assignment has a similar probability with and without replacement

Randomization designs

Classic Randomization Designs

- Classic randomization designs can be characterized by [restrictions on \$\mathbb{W}^+\$](#)

Type of Experiment and Design	Number of Possible Assignments Cardinality of $\mathbb{W}^+$	Number of Units (N) in Sample			
		4	8	16	32
Bernoulli trial	2^N	16	256	65,536	4.2×10^9
Completely randomized experiment	$\binom{N}{N/2}$	6	70	12,870	0.6×10^9
Stratified randomized experiment	$\left(\binom{N/2}{N/4}\right)^2$	4	36	4,900	0.2×10^9
Paired randomized experiment	$2^{N/2}$	4	16	256	65,536

Bernoulli Trials (Coin Tossing)

- $p_i = e = q = 0.5, \mathbb{W}^+ = \{0, 1\}^N, \dim(\mathbb{W}^+) = 2^N$

$$P(\mathbf{W}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = 0.5^N$$

- More generally, with $q \neq 0.5$

$$P(\mathbf{W}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = q^{N_t} (1 - q)^{N_c}$$

- No way to ensure “enough” treated and control units under each assignment

Completely Randomized Experiments

- Draw N_t units at random, such that $1 \leq N_t \leq N - 1$
- $q = \frac{N_t}{N}$, and $\dim(\mathbb{W}^+) = \binom{N}{N_t}$
- The set of possible assignment vectors is

$$\mathbb{W}^+ = \left\{ \mathbf{w} \in \mathbb{W} \mid \sum_{i=1}^N W_i = N_t \right\}$$

- Possible **issue with covariate unbalancedness** after treatment assignment

Stratified Randomized Experiments

- The population of units is first partitioned into **blocks or strata** $B_i = B(\mathbf{X}_i)$
- Within each block, we conduct a completely randomized experiment
- The set of possible assignment vectors is

$$\mathbb{W}^+ = \left\{ \mathbf{w} \in \mathbb{W} \mid \sum_{i: B_i=j}^N W_i = N_t(j) \right\}$$

- **More precise inference** as units of same type are in different treat arms

Paired Randomized Experiments

- As many units as treatments within each block
- $N(j) = 2$ and $N_t(j) = 1$ for $j = 1, \dots, N/2$, so that

$$\mathbb{W}^+ = \left\{ \mathbf{w} \in \mathbb{W} \mid \sum_{i: B_i = j}^N W_i = 1 \right\}.$$

- Useful design when N is small and/or J is large

Other Randomization Designs (Popular in Econ)

- Clustered randomized experiments
- Partial population designs
- Randomized saturation designs
- Adaptive randomized experiments

Clustered Randomized Experiments

- Clusters are defined by partitioning the covariate space $G_{ig} = G(\mathbf{X}_i)$
- $\bar{W}_g = \sum_{i:G_{ig}=1} \frac{W_i}{N_g} \in \{0, 1\}$ is the average value of W_i for units in cluster g
- The assignment mechanisms concerns groups of units (clusters)

$$\mathbb{W}^+ = \left\{ \mathbf{W} \in \mathbb{W} \mid \sum_{g=1}^G \bar{W}_g = G_t \right\}$$

- Relax SUTVA within clusters, but maintain it across clusters

Partial Population Designs

- Within-cluster (non-random) program assignment $E_{ig} = \{0, 1\}$
- Write potential outcomes as $Y_i(W_g, E_{ig})$

$$\begin{aligned}ATE &= \mathbb{E}(Y_i(1, 1) - Y_i(0, 1)) = \\&= \mathbb{E}(Y_i | W_g = 1, E_{ig} = 1) - \mathbb{E}(Y_i | W_g = 0, E_{ig} = 1) \\ITE &= \mathbb{E}(Y_i(1, 0) - Y_i(0, 0)) = \\&= \mathbb{E}(Y_i | W_g = 1, E_{ig} = 0) - \mathbb{E}(Y_i | W_g = 0, E_{ig} = 0)\end{aligned}$$

- ATE and ITE depend on $S_g = \sum_{i \in g} E_{i,g}$, which is endogenous

Randomized Saturation Design (Two-Step Randomization)

- Assign each cluster to a treatment saturation, $S_g = \sum_{i \in g} W_{ig} \in [0, 1]$
- Assign each individual to a treatment status $W_{ig} = \{0, 1\}$ according to S_g
- Write potential outcomes as $Y_i(W_{ig}, S_g)$

$$\begin{aligned}ATE(s) &= \mathbb{E}(Y_i(1, s) - Y_i(0, 0)) = \\&\quad + \mathbb{E}(Y_i | W_{ig} = 1, S_g = s) - \mathbb{E}(Y_i | W_{ig} = 1, S_g = 0) \\ITE(s) &= \mathbb{E}(Y_i(0, s) - Y_i(0, 0)) = \\&\quad = \mathbb{E}(Y_i | W_{ig} = 0, S_g = s) - \mathbb{E}(Y_i | W_{ig} = 0, S_g = 0)\end{aligned}$$

- Optimal combination of clustered and stratified designs

Adaptive Randomized Experiments

- Repeated cross-sections: $(Y_{it}^1, \dots, Y_{it}^k)$ are i.i.d. across both i and t
- Treatment $W \in \{1, \dots, k\}$, potential outcomes $Y(w)$
- Decide what share p_t^w will be assigned to treatment W in time t
- **Posterior beliefs** P_t over $Y(w)$ (e.g. thompson sampling):

$$p_t^w = P_t \left(w = \underset{w'}{\operatorname{argmax}} Y(w') \right)$$

Stratified randomized experiments

The Structure of Stratified Randomized Experiments

- Let J be the number of **strata/blocks**, and $N(j), N_c(j), N_t(j)$
- Let $S_i \in \{1, \dots, J\}$ be the stratum for unit i
- Let $B_i(j) = \mathbf{1}_{S_i=j}$ be the **stratum indicator** for unit i
- The assignment mechanism is

$$P(\mathbf{W} | \mathbf{B}, \mathbf{Y}(0), \mathbf{Y}(1)) = \prod_{j=1}^J \binom{N(j)}{N_t(j)}^{-1} \text{ for } \mathbf{W} \in \mathbb{W}^+$$

$$\Rightarrow \mathbb{W}^+ = \{\mathbf{W} \in \mathbb{W} \mid \sum_{i=1}^N B_i(j) \cdot W_i = N_t(j) \text{ for } j = 1, \dots, J\}$$

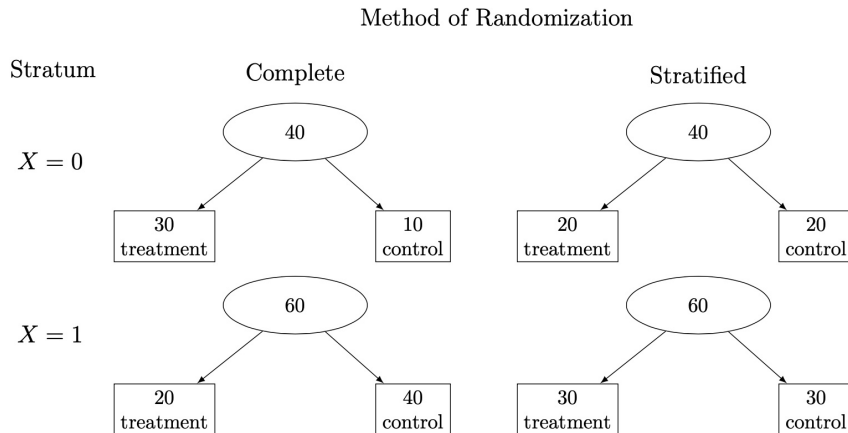
Example: Tennessee Project Star

School/ Stratum	No. of Classes	Regular Classes ($W_i = 0$)	Small Classes ($W_i = 1$)
1	4	-0.197, 0.236	0.165, 0.321
2	4	0.117, 1.190	0.918, -0.202
3	5	-0.496, 0.225	0.341, 0.561, -0.059
4	4	-1.104, -0.956	-0.024, -0.450
5	4	-0.126, 0.106	-0.258, -0.083
6	4	-0.597, -0.495	1.151, 0.707
7	4	0.685, 0.270	0.077, 0.371
8	6	-0.934, -0.633	-0.870, -0.496, -0.444, 0.392
9	4	-0.891, -0.856	-0.568, -1.189
10	4	-0.473, -0.807	-0.727, -0.580
11	4	-0.383, 0.313	-0.533, 0.458
12	5	0.474, 0.140	1.001, 0.102, 0.484
13	4	0.205, 0.296	0.855, 0.509
14	4	0.742, 0.175	0.618, 0.978
15	4	-0.434, -0.293	-0.545, 0.234
16	4	0.355, -0.130	-0.240, -0.150
Average		-0.13	0.09
(S.D.)		(0.56)	(0.61)

The Benefits of Stratification

- DGP with $i = 1, \dots, N$, one covariate $X_i \in \{0, 1\}$ and $\pi = \sum_i X_i / N$
- Consider the following two assignment mechanisms for W_i
 - ⇒ Complete randomization (CR): $N_t = qN$ and $N_c = (1 - q)N$
 - ⇒ Stratified randomization (SBR): $N_{t,X=1} = \pi qN$, $N_{t,X=0} = (1 - \pi)qN$ and $N_{c,X=1} = \pi(1 - q)N$, $N_{c,X=0} = (1 - \pi)(1 - q)N$
- SBR reproduces the composition in the overall sample in both $W = 1$ and $W = 0$
- CR generates possible imbalance of W for $X = 1$ and $X = 0$ and higher variance of ATE

Complete Randomization Vs. Stratified: Example with $N = 100$



Randomization Inference for Stratified Experiments

- Sharp null hypothesis

$$H_0 : Y_i(1) = Y_i(0) \forall i = 1, 2, \dots, N.$$

- Define average observed outcomes in stratum j as

$$\bar{Y}_t^{\text{obs}}(j) = \frac{1}{N_t(j)} \sum_{i:S_i=j} W_i Y_i^{\text{obs}}$$

$$\bar{Y}_c^{\text{obs}}(j) = \frac{1}{N_c(j)} \sum_{i:S_i=j} (1 - W_i) Y_i^{\text{obs}}$$

- Strata-level propensity score is

$$e(j) = \frac{N_t(j)}{N(j)}$$

Test Statistics

- Within-stratum test statistic

$$T^{\text{dif}}(j) = |\bar{Y}_t^{\text{obs}}(j) - \bar{Y}_c^{\text{obs}}(j)|$$

⇒ Not very informative as we are interested in treatment effects across all strata

- Linear combination of the within-stratum statistics

$$T^{\text{dif}, \lambda_{RSS}} = \left| \sum_{j=1}^J \frac{N_j}{N} \left(\bar{Y}_t^{\text{obs}}(j) - \bar{Y}_c^{\text{obs}}(j) \right) \right|$$

⇒ Need $e(j)$ to vary across j for the test to have power over $T^{\text{dif}} = |\bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}}|$

Randomization Inference of the Tennessee Project Star

- $B_i(j), i = 1, \dots, 68$ (class-level data)

$$H_0 : Y_i(1) = Y_i(0) \forall i = 1, 2, \dots, 68.$$

- Total number of **possible assignments** is $6^{13} \times 10^2 \times 15$

⇒ 13 Schools with two classes in each group: $\binom{4}{2} = 6$

⇒ 2 Schools with three small classes and two regular classes: $\binom{5}{2} = 10$

⇒ 1 School with four small classes and two regular classes: $\binom{6}{2} = 15$

$$\Rightarrow T^{\text{dif}} = |\bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}}| = 0.224, \text{ with } p = 0.034$$

$$\Rightarrow T^{\text{dif}, \lambda_{RSS}} = \left| \sum_{j=1}^J \frac{N_j}{N} (\bar{Y}_t^{\text{obs}}(j) - \bar{Y}_c^{\text{obs}}(j)) \right| = 0.241, \text{ with } p = 0.023$$

Regression Analysis

$$Y_i^{\text{obs}} = \tau W_i + \sum_{j=1}^J \beta(j) B_i(j) + \epsilon_i$$

- $\hat{\tau}^{\text{ols}}$ is not a consistent estimator of ATE if $\tau(j) \neq \tau(j') \forall j \neq j'$

$$WATE = \frac{\sum_{j=1}^J \omega(j) \tau(j)}{\sum_{j=1}^J \omega(j)}$$

- $\omega(j) = \frac{N_j}{N} \frac{N_t(j)}{N(j)} \frac{N(j) - N_t(j)}{N(j)} = q(j) e(j) (1 - e(j))$

Asymptotic Inference

- The asymptotic variance of the WATE is

$$\mathbb{V}^{\text{strat}} = \frac{\sum_{i=1}^N \epsilon_i^2 \cdot \left(W_i - \sum_{j=1}^J q(j) B_i(j) \right)^2}{\left(\sum_{j=1}^J \omega(j) \right)^2}$$

- Variance weights more those $\hat{\tau}(j)$ that are more precisely estimated

$$\mathbb{V}^j = \frac{\sigma^2/N}{q(j)e(j)(1-e(j))}$$

Fully-interacted Model

$$Y_i^{\text{obs}} = \tau W_i \frac{B_i(j)}{N(j)/N} + \sum_{j=1}^J \beta(j) B_i(j) + \sum_{j=1}^{J-1} \gamma(j) W_i \left(B_i(j) - B_i(J) \frac{N(j)}{N(J)} \right) + \epsilon_i$$

- In this case, $\hat{\tau}^{\text{ols}}$ is a consistent estimator of ATE
- With asymptotic variance

$$\mathbb{V}^{\text{strat,inter}} = \sum_{i=1}^N q(j)^2 \cdot \left(\frac{\sigma_c^2(j)}{q(j)(1-e(j))} + \frac{\sigma_t^2(j)}{q(j)e(j)} \right)$$

- In general, $\mathbb{V}^{\text{strat,inter}} > \mathbb{V}^{\text{strat}}$

Regression Analysis of the Tennessee Project Star

- The point estimate of τ in the standard model is
 - $\hat{\tau}^{\text{ols}} = 0.238$ ($\widehat{s.e.} = 0.103$)
- The point estimate of τ in the fully-interacted model is
 - $\hat{\tau}^{\text{ols,inter}} = 0.241$ ($\widehat{s.e.} = 0.095$)
- Limited heterogeneity in the treatment effects across strata

Clustered Randomized Experiments

The Structure of Clustered Experiments

- Let G_{ig} be a binary indicator that unit i belongs to cluster $g = 1, \dots, G$
- $N_g = \sum_{i=1}^N G_{ig}$, so that N_g/N is the share of cluster g in the sample
- $\bar{W}_g \in \{0, 1\}$ is the **treatment assignment for all units** in cluster g
- The assignment mechanism is

$$P(\mathbf{W}, \mathbf{Y}(0), \mathbf{Y}(1), \mathbf{X}) = \binom{G}{G_t}^{-1}$$

$$\Rightarrow \mathbb{W}^+ = \{\mathbf{W} \in \mathbb{W} \mid \sum_{g=1}^G \bar{W}_g = G_t\}$$

Example: The *Progresa* Program (Again!)

- Clustered RCT during the roll-out of the program in rural areas
 - ⇒ 506 villages among those eligible to receive the program
 - ⇒ 320 early treatment and 186 late treatment (control)
- Individual/HH level data for both **eligible and non-eligible HHs** in each village
 - ⇒ ~30,000 program eligible children
 - ⇒ About 50-100 HHs per village

Estimands

- The choice of estimand depends on the **unit of analysis**
- **Unit-level**: a natural estimand is the usual ATE

$$\tau = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))$$

- **Cluster-level**: weighted average of the unit-level treatment effect

$$\tau^{\text{clust}} = \frac{1}{G} \sum_{g=1}^G \tau_g, \quad \text{where } \tau_g = \frac{1}{N_g} \sum_{i:G_{ig}=1}^N (Y_i(1) - Y_i(0))$$

Unit-level Vs. Cluster-level

- Cluster-level analysis is more directly linked to the randomization framework
 - ⇒ **Differently-sized clusters**, such as states or towns
 - ⇒ Many units will be in the same treatment group so unit-level inference is tricky
- Unit-level incorporates individual-level covariates, which improve efficiency
 - ⇒ **More homogenous clusters**, such as schools or classrooms

Randomization Inference

- The usual statistic for **unit-level** analysis

$$T^{\text{dif}} = | \bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}} |$$

- The equivalent statistic for **cluster-level** analysis

$$T^{\text{clust}} = \left| \frac{1}{G_t} \sum_{g: \bar{W}_g=1} \bar{Y}_g^{\text{obs}} - \frac{1}{G_c} \sum_{g: \bar{W}_g=0} \bar{Y}_g^{\text{obs}} \right|$$

Randomization Inference of *Progresa*

- Children-level analysis on school enrollment (pre-program year 1997)

$$T^{\text{dif}} = 0.0075, \quad p\text{-value} = 0.400$$

- Children-level analysis on school enrollment (program year 1998)

$$T^{\text{dif}} = 0.0388, \quad p\text{-value} < 0.001$$

- Village-level analysis on school enrollment (program year 1998)

$$T^{\text{clust}} = 0.0234, \quad p\text{-value} = 0.0120$$

Regression Methods

- In **unit-level** analysis, we estimate the following regression

$$Y_i^{\text{obs}} = \alpha + \tau W_i + \bar{X}'\gamma + \epsilon_i$$

- $\hat{\tau}^{\text{ols}}$ consistently estimates ATE (with centered covariates)
- Use Liang-Zeger **clustered Var-Cov**:

$$\mathbb{V}_{\text{clust}} = \frac{\sum_{g=1}^G \left(\sum_{i: G_{ig}=1} \epsilon_i^2 \cdot (W_i - \bar{W})^2 \right)}{\left(\sum_{i=1}^N (W_i - \bar{W})^2 \right)^2}$$

Regression Analysis: Cluster-Level

- In **cluster-level** analysis, consider the following regression

$$\overline{Y}_g^{\text{obs}} = \alpha + \tau \overline{W}_g + \eta_g$$

- $\hat{\tau}^{\text{ols}}$ consistently estimates ATE
- The sampling variance of τ^{ols} is the usual one

$$\mathbb{V} = \frac{\sum_{g=1}^G \eta_g^2}{\sum_{g=1}^G (\overline{W}_g - \overline{W})^2} = \sigma^2 \left\{ \frac{1}{G_t} + \frac{1}{G_c} \right\}$$

Regression Analysis of *Progres*

- Children-level analysis on school enrollment (*pre-program* year 1997)

$$\hat{\tau}^{\text{ols}} = 0.0075 \ (\widehat{s.e.} = 0.0091)$$

- Children-level analysis on school enrollment (*program* year 1998)

$$\hat{\tau}^{\text{ols}} = 0.0388 \ (\widehat{s.e.} = 0.0104)$$

- Village-level analysis on school enrollment (*program* year 1998)

$$\hat{\tau}^{\text{ols}} = 0.0234 \ (\widehat{s.e.} = 0.0092)$$

Dynamic Latent Factor Models

Human Capital Accumulation in the Early Years

- The accumulation of human capital is a complex process
 - ⇒ Human capital is **multi-dimensional** (cognitive, non-cognitive, health,...)
 - ⇒ Skill formation is a **dynamic** process (skills beget skills)
 - ⇒ Both **genetic and environmental factors** are important inputs
- ⇒ The **early years** is a particularly salient period for policy
 - ⇒ Human capital is malleable (and vulnerable)
 - ⇒ There is **no equity-efficiency trade-off** for early investment

The Production Function of Human Capital

- Consider the following **recursive formulation** for skill accumulation

$$\mathbf{H}_{t+1} = g_t(\mathbf{H}_t, \mathbf{X}_t, \mathbf{I}_{t+1}, u_{t+1})$$

- $\Rightarrow \mathbf{H}_t$ is human capital (including cognition, socio-emotional, and health)
- $\Rightarrow \mathbf{X}_t$ are parental skills and other background variables
- $\Rightarrow \mathbf{I}_{t+1}$ are parental investments (including materials and time) between t and $t + 1$
- $\Rightarrow$ Stocks of **current period skills produce next period skills**

Self-Productivity and Dynamic Complementarities

- Self-productivity arises when

$$\frac{\partial g_t}{\partial \mathbf{H}_t} > 0$$

- Period t stocks of skills promote the acquisition of skills by making investment more productive

$$\frac{\partial^2 g_t}{\partial I_t \partial \mathbf{H}_t} > 0$$

- Explains why returns to education are higher later in child life cycle for more able children

Three Challenges in the Estimation of HK Production Functions

- Anchoring

⇒ Need to normalize the scales of each latent factor to measures with cardinal interpretation

- Measurement error

⇒ Use dynamic latent factors models where skills are proxied by vectors of measurements

- Endogeneity

⇒ Observed inputs likely correlate with unobserved inputs

Measurement System

- For each $k \in \{c, n, m, f, \dots\}$ and period t we have measures such that:

$$m_{k,t,j} = \alpha_{k,t,j} \theta_{k,t} + \epsilon_{k,t,j}$$

- $\Rightarrow m_{k,t,j}$ is the j^{th} measurement of skill $\theta_{k,t}$
 - $\Rightarrow \alpha_{k,t,j}$ is a factor loading
 - $\Rightarrow \epsilon_{k,t,j}$ is the measurement error contained in $m_{k,t,j}$
- With **two measures per factor**, the distributions of factors and measurement error are identified
 - $\Rightarrow$ Set the factor scale: $\alpha_{k,t,1} = 1$ + factor location: $E(\theta_{k,t}) = 0, \forall t = 1, \dots, T$
 - $\Rightarrow$ Need $J = 2M + 1$ measures to identify factor loadings for M factors

Identification: Factor Loadings

- Assume that the $\epsilon_{k,t,j}$ are uncorrelated across t , then

$$\text{Cov}(m_{k,t,1}, m_{k,t+1,1}) = \text{Cov}(\theta_{k,t}, \theta_{k,t+1})$$

$$\text{Cov}(m_{k,t,2}, m_{k,t+1,1}) = \alpha_{k,t,2} \text{Cov}(\theta_{k,t}, \theta_{k,t+1})$$

- Hence, we can identify the **factor loading**

$$\frac{\text{Cov}(m_{k,t,1}, m_{k,t+1,1})}{\text{Cov}(m_{k,t,2}, m_{k,t+1,1})} = \alpha_{k,t,2}$$

- if $J > 2M + 1$ one can also **allow for serial correlation** in the $\epsilon_{k,t,j}$

Identification: Distribution of Latent Variables

- Once $\alpha_{k,t,j}$ are identified, we can re-write the measurement system as

$$\frac{m_{k,t,j}}{\alpha_{k,t,j}} = \theta_{k,t} + \frac{\epsilon_{k,t,j}}{\alpha_{k,t,j}}$$

- The **signal is common to multiple measurements**, but the noise is not
- To identify the **joint distribution of $\{\theta_{k,t}\}_{t=1}^T$** , assume that the $\epsilon_{k,t,j}$ (these can be relaxed):
 - ⇒ Are uncorrelated across measures conditional on θ
 - ⇒ Are independent from factors

The Investment Problem

- Parents choose investments by solving

$$\begin{aligned} \max_{\{C_t, I_{t+1}\}} & U(C_t, \mathbf{H}_{t+1}) \\ \text{s.t. } & C_t + \mathbf{P}_t^I I_{t+1} = Y_t \\ & \mathbf{H}_{t+1} = g_t(\mathbf{H}_t, \mathbf{X}_t, \mathbf{I}_{t+1}, u_{t+1}) \end{aligned}$$

- Investment could be in **commodities (i.e., material)** or in **time**
- This can be generalized (add parental labor supply decisions + other inter-temporal dimensions)
- **Parents choose investment** in reaction to shocks, and so $E(u_{t+1} | I_{t+1}) \neq 0$

Endogeneity of Investment

- Posit a reduced-form of the investment function

$$I_{t+1} = f_t(Y_t, P_t, \mathbf{X}_t, e_{t+1})$$

- Y_t and P_t are **excluded from the production function**
- One can estimate the investment equation jointly with the production function
- Or use a **control function approach** to estimate the role of investments in the production function
 - ⇒ Plug estimated residuals $\hat{e}_{t+1}$ in the production function, so that $E(u_{t+1}|I_{t+1}, \hat{e}_{t+1}) = 0$

Estimation

- Estimation can be performed in **three-steps**
 - ⇒ Estimate covariance structure and factor loadings of the latent factors
 - ⇒ Match the moments of the observed measures to the moments imposed by the factor structure
 - ⇒ Use the predicted scores to estimate the parental investment and production function equations
- Inference can be done by using appropriate **bootstrapping**
 - ⇒ Takes into account estimation error + simulation error

The production of human capital in the early years

Attanasio-Cattan-Fitzsimons-Meghir-Rubio (2020, AER)

- Adapt a prominent **ECD program** to the Colombian context
 - ⇒ Children 12-24 months living in families targeted by a cash transfer program
 - ⇒ Cheap and scalable intervention
- **Estimate production functions** for child cognitive and socio-emotional skills
 - ⇒ Quantify by how much crowd-in of parental investments contributed to improving child outcomes
 - ⇒ The direct effect of the intervention is small and insignificant

Research Design

- Clustered RCT, stratified by region
 - ⇒ 1,429 children in 96 municipalities in central Colombia
- Municipalities were randomized into 4 groups (cross-cutting or factorial design)
 - ⇒ Control
 - ⇒ Stimulation (home visits)
 - ⇒ Supplementation (micronutrients)
 - ⇒ Stimulation + Supplementation

Rich Data Collected on Children and Parents

- A battery of **psychometrics on children development** (baseline and follow-up)
 - ⇒ Motor and cognitive skills: Bayley test
 - ⇒ Socio-emotional skills: Bates temperament test
 - ⇒ Language development: MacArthur-Bates test
 - ⇒ Height, weight, hemoglobin, and morbidity
 - ⇒ Food intakes
 - ⇒ Child care arrangements and time use
- Mothers and families
- Home visitors
- Qualitative instruments

Treatment Impacts of Stimulation on Measures and Factors

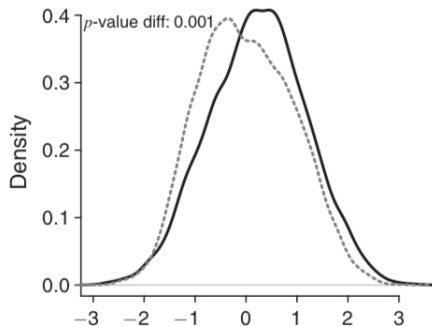
	Treatment effect		
	Point estimate	SE	Sample size
<i>Panel A. Child's cognitive skills at follow-up</i>			
Bayley: Cognitive	0.250	(0.063)	1,264
Bayley: Receptive language	0.175	(0.063)	1,264
Bayley: Expressive language	0.032	(0.062)	1,263
Bayley: Fine motor	0.072	(0.060)	1,262
MacArthur: Words the child can say	0.092	(0.064)	1,322
MacArthur: Complex phrases the child can say	0.058	(0.055)	1,322
Cognitive factor	0.115	(0.051)	
<i>Panel B. Child's socio-emotional skills at follow-up</i>			
ICQ: Difficult (-)	0.074	(0.045)	1,326
ICQ: Unsociable (-)	0.041	(0.054)	1,326
ICQ: Unstoppable (-)	0.032	(0.054)	1,326
ECBQ: Inhibitory control	-0.003	(0.058)	1,323
ECBQ: Attentional focusing	0.069	(0.049)	1,323
Socio-emotional factor	0.087	(0.044)	
<i>Panel C. Material investments at follow-up</i>			
FCI: Number of types of play materials	0.215	(0.064)	1,326
FCI: Number of coloring and drawing books	-0.133	(0.056)	1,326
FCI: Number of toys to learn movement	-0.048	(0.065)	1,326
FCI: Number of toys to learn shapes	0.416	(0.088)	1,326
FCI: Number of shop-bought toys	0.024	(0.061)	1,326
Material investment factor	0.227	(0.069)	
<i>Panel D. Time investments at follow-up</i>			
FCI: Number of types of play activities in last 3 days	0.277	(0.050)	1,326
FCI: Number of times told a story to child in last 3 days	0.138	(0.060)	1,326
FCI: Number of times read to child in last 3 days	0.362	(0.062)	1,326
FCI: Number of times played with toys in last 3 days	0.175	(0.060)	1,326
FCI: Number of times named things to child in last 3 days	0.137	(0.048)	1,326
Time investment factor	0.302	(0.068)	

Experimental Results

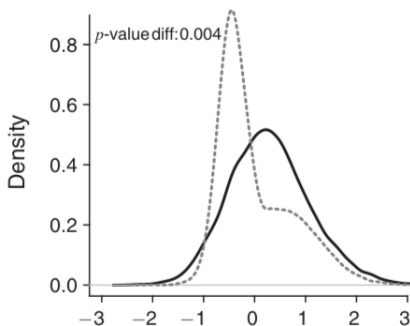
- Stimulation improved cognitive scores by 0.25σ + receptive language by 0.18σ
- Also positive (noisy) effects on socio-emotional skills
- Micronutrient supplementation had no significant effect on any outcome
- No effect of the intervention on height, weight, or hemoglobin levels

A First Hint at the Mechanisms: Parental Investment

Panel E. Material investments, follow-up



Panel F. Time investments, follow-up



— Treated ---- Control

What is the Rationale for the Model?

- Intervention changes human capital through **three different channels**
 - ⇒ Total factor productivity
 - ⇒ The productivity of the individual inputs
 - ⇒ The distribution of inputs (and in particular parental investment)
- Jointly estimate the **distribution of inputs and the production function**
- Unpack the role of these different hypotheses

Measurement Equations

$$m_{j,d}^{\theta} = \mu_{j,d}^k + \alpha_j^k \ln \theta_d^k + \epsilon_j^k$$

- Treated and control children are allowed to have different factor distributions
- Latent factors are distributed as a mixture of two normal distributions
- Measurement errors terms are jointly normal
- Likelihood function for each treatment/control groups writes as

$$L_i(m) = \int_{\theta} f(m_i|\theta) [\tau \phi_A(\theta) + (1 - \tau) \phi_b(\theta)] d\theta$$

- Anchor cognitive factor to the Bayley cognitive score (the number of tasks completed correctly)

The Production Function for Human Capital

- Cobb-Douglas specification for the technology of skill formation

$$\ln(\theta_{i,d,t+1}^k) = A_d^k + \gamma_{1,d}^k \ln(\theta_{i,t}^C) + \gamma_{2,d}^k \ln(\theta_{i,t}^S) + \gamma_{3,d}^k \ln(P_{i,t}^C) + \gamma_{4,d}^k \ln(P_{i,t}^S) \\ + \gamma_{5,d}^k \ln(I_{i,t+1}^M) + \gamma_{6,d}^k \ln(I_{i,t+1}^T) + \gamma_{7,d}^k \ln(n_{i,t}) + \eta_{i,t+1}^k$$

- A_d^k captures the direct effect of the home-visitor during the weekly visit

⇒ All parameters are skill-specific and depend on treatment status

Parental Investments

- Same log-linear specification for investment equation ($\tau = \{M, T\}$)

$$\begin{aligned} \ln(I_{i,t+1}^\tau) = & \lambda_{0,d}^\tau + \lambda_{1,d}^\tau \ln(\theta_{i,t}^C) + \lambda_{2,d}^\tau \ln(\theta_{i,t}^S) + \lambda_{3,d}^\tau \ln(P_i^C) \\ & + \lambda_{4,d}^\tau \ln(P_i^S) + \lambda_{5,d}^\tau \ln(n_{i,t}) + \lambda_{6,d}^\tau Z_{i,t} + u_{i,t+1} \end{aligned}$$

- Z_t are instruments
- All parameters are allowed to vary with the treatment (although de facto they don't)
- **Control function** to account for correlation between shocks and investments

⇒ Include residuals $\hat{u}_{i,t+1}$ as additional regressor in the production function

Evidence Supporting the Exclusion Restrictions

TABLE 3—BALANCE TEST FOR THE INSTRUMENTAL VARIABLES

	log toy price	log food price	Conflict
Mother's cognitive skill	0.021 (0.009)	0.007 (0.007)	−0.004 (0.003)
Mother's socio-emotional skill	−0.006 (0.008)	−0.014 (0.006)	0.005 (0.005)
Mother is married	0.013 (0.016)	0.030 (0.012)	0.017 (0.008)
Wealth index	−0.003 (0.007)	0.002 (0.006)	0.002 (0.004)
Terrorism	0.015 (0.014)	−0.012 (0.013)	0.005 (0.009)
Constant	8.025 (0.028)	8.062 (0.017)	0.049 (0.007)
Observations	1,010	1,023	1,023
R^2	0.011	0.021	0.017
F -statistic	1.656	2.147	1.438
F -test p -value	0.154	0.0671	0.219

Notes: All right-hand-side variables measured at baseline. Asymptotic standard errors in parentheses allowing for clustering at the municipality level. *Terrorism* is the number of terrorist attacks between conception of child and baseline. *Conflict* is the number of conflicts against civil population divided by the municipality population (in thousands) when the mother herself was a child.

Estimation

- Estimate the **covariance of the factors** based on covariance of the observed measures
- Use these estimates to **predict factor scores**: $\hat{\theta}_{i,t}^k, \hat{\theta}_{i,t}^k$
- Use predicted scores as data to **estimate the parental investments and production function**
- Compute confidence intervals for test statistics using the **block bootstrap**

Investment Functions

TABLE 5—ESTIMATES OF THE MATERIAL AND TIME INVESTMENT EQUATIONS

	Instruments: prices and conflict		Instruments: prices only
	Material investment	Time investment	Material investment
Intercept	−0.015 [−0.114, 0.078]	0.001 [−0.089, 0.089]	−0.013 [−0.11, 0.078]
Treatment	0.209 [0.038, 0.365]	0.318 [0.155, 0.48]	0.204 [0.037, 0.362]
log child's cognitive skill (t)	0.130 [0.016, 0.246]	0.068 [−0.044, 0.18]	0.132 [0.017, 0.25]
log child's socio-emotional skill (t)	−0.028 [−0.133, 0.087]	0.027 [−0.083, 0.145]	−0.030 [−0.131, 0.088]
log mother's cognitive skill	0.748 [0.582, 0.939]	0.349 [0.162, 0.498]	0.750 [0.583, 0.943]
log mother's socio-emotional skill	0.069 [−0.008, 0.139]	0.022 [−0.06, 0.108]	0.068 [−0.008, 0.139]
log number of children	−0.129 [−0.18, −0.077]	−0.128 [−0.186, −0.072]	−0.128 [−0.18, −0.078]
log toy price	−0.096 [−0.168, −0.027]	−0.020 [−0.085, 0.037]	−0.094 [−0.163, −0.026]
log food price	0.091 [0.006, 0.178]	0.042 [−0.026, 0.121]	0.091 [0.006, 0.178]
Maternal childhood exposure to conflict	−0.009 [−0.08, 0.063]	−0.089 [−0.139, −0.032]	

Production Function of Cognitive Skills

TABLE 6—ESTIMATES OF THE PRODUCTION FUNCTION FOR COGNITIVE SKILLS

Instruments:	OLS	IV			
	(1)	Prices, conflict (2)	Prices, conflict, treatment (3)	Prices (4)	Prices and treatment (5)
Intercept	−0.018 [−0.094, 0.053]	−0.019 [−0.111, 0.079]	0.007 [−0.089, 0.104]	0.003 [−0.091, 0.072]	−0.009 [−0.083, 0.058]
Treatment	0.083 [−0.025, 0.192]	0.049 [−0.12, 0.391]		−0.028 [−0.186, 0.156]	
log child's cognitive skill (<i>t</i>)	0.675 [0.589, 0.77]	0.648 [0.544, 0.795]	0.638 [0.522, 0.747]	0.626 [0.525, 0.746]	0.631 [0.533, 0.739]
log child's socio-emotional skill (<i>t</i>)	0.001 [−0.091, 0.087]	0.012 [−0.098, 0.143]	0.019 [−0.094, 0.14]	0.015 [−0.084, 0.126]	0.02 [−0.079, 0.127]
log mother's cognitive skill	0.213 [0.089, 0.35]	−0.075 [−0.456, 0.5]	−0.173 [−0.538, 0.201]	−0.102 [−0.495, 0.291]	−0.094 [−0.45, 0.21]
log mother's socio-emotional skill	0.103 [0.031, 0.173]	0.084 [−0.03, 0.163]	0.063 [−0.035, 0.151]	0.06 [−0.019, 0.152]	0.074 [−0.026, 0.155]
log number of children	0.042 [−0.01, 0.092]	0.085 [−0.07, 0.154]	0.084 [0.011, 0.163]	0.089 [0.002, 0.176]	0.086 [0.023, 0.164]
log material investment	0.088 [0.016, 0.157]	0.594 [0.025, 1.179]	0.784 [0.204, 1.383]	0.542 [0.041, 0.996]	0.516 [0.195, 0.946]
log time investment	0.038 [−0.051, 0.129]	−0.171 [−1.198, 0.312]	−0.311 [−0.985, 0.217]		
Goodness-of-fit: Gap in output between treated and control					
Measured in the data	0.115	0.115	0.115	0.115	0.115
Predicted by the model	0.114	0.132	0.084	0.095	0.117

Production Function of Socio-emotional Skills

TABLE 7—ESTIMATES OF THE PRODUCTION FUNCTION FOR SOCIO-EMOTIONAL SKILLS

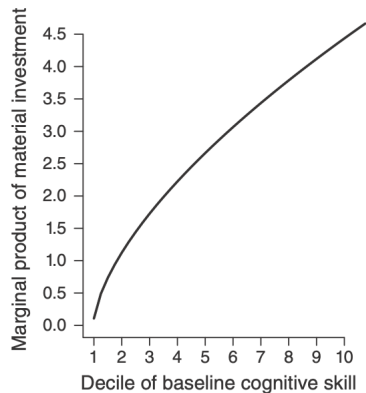
Instruments:	OLS	IV			
	(1)	Prices, conflict (2)	Prices, conflict (3)	Prices, conflict, treatment (4)	Prices, conflict, treatment (5)
Intercept	−0.009 [−0.08, 0.063]	−0.02 [−0.087, 0.064]	−0.006 [−0.084, 0.066]	−0.02 [−0.096, 0.058]	−0.022 [−0.089, 0.051]
Treatment	−0.011 [−0.124, 0.093]	−0.071 [−0.318, 0.166]	−0.116 [−0.31, 0.115]		
log child's cognitive skill (<i>t</i>)	0.106 [0.018, 0.192]	0.074 [−0.019, 0.21]	0.09 [−0.017, 0.194]	0.094 [0.003, 0.22]	0.099 [0.002, 0.193]
log child's socio-emotional skill (<i>t</i>)	0.522 [0.403, 0.659]	0.513 [0.374, 0.672]	0.499 [0.389, 0.663]	0.516 [0.387, 0.669]	0.512 [0.403, 0.656]
log mother's cognitive skill	−0.077 [−0.217, 0.049]	−0.146 [−0.443, 0.297]	−0.15 [−0.349, 0.084]	−0.018 [−0.315, 0.328]	−0.083 [−0.231, 0.078]
log mother's socio-emotional skill	0.037 [−0.058, 0.119]	0.035 [−0.062, 0.141]	0.048 [−0.053, 0.135]	0.049 [−0.054, 0.134]	0.043 [−0.042, 0.126]
log of number of children	0.099 [0.047, 0.153]	0.127 [0.017, 0.236]	0.133 [0.034, 0.223]	0.101 [0.028, 0.168]	0.101 [0.036, 0.163]
log material investment	0.154 [0.06, 0.256]	0.015 [−0.621, 0.428]		−0.144 [−0.688, 0.34]	
log time investment	0.109 [−0.006, 0.213]	0.487 [−0.177, 1.258]	0.549 [−0.022, 1.147]	0.448 [−0.125, 1.057]	0.324 [0.025, 0.691]
Goodness-of-fit: Gap in output between treated and control					
Measured in the data	0.087	0.087	0.087	0.087	0.087
Predicted by the model	0.057	0.080	0.050	0.103	0.098

Interpreting the Impact of the Intervention

- Cognitive (Socio-emotional) development is mainly determined by
 - ⇒ Initial cognition (measured at ages 1-2)
 - ⇒ Stimulation provided by older siblings
 - ⇒ **Material (time) investments** provided by parents
- The main channel underlying the impact of the intervention is a **shift in parental investments**

Complementarities between Inputs

Panel A. Production function for cognitive skills



Panel B. Production function for socio-emotional skills

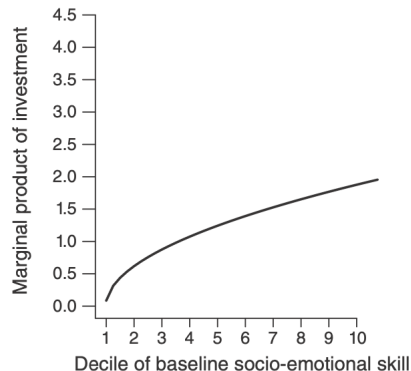


FIGURE 2. COMPLEMENTARITY BETWEEN INVESTMENTS AND BASELINE SKILLS

Implications for longer-term outcomes

- Parental crowding-in suggests treatment will diverge in the medium/long run
- But also possible fade-out due to mean reversion
- Follow-up data (2 years after) suggests **partial fade out**
- How can we permanently **improve parental practices** ?

RCT $\Leftrightarrow$ Economic Model

- Model gives a lens to **interpret how the intervention** affected child development
- Sometimes the **experimental variation is what you want to explain**
- Need additional sources of variation beyond the experiment